Automating Kinship Recognition

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1 Introduction

Family photographs serve as timeless repositories of memories, capturing moments that connect generations. Yet, as time passes, the identities and relationships of the people in these images often fade into obscurity. Many individuals find themselves staring at old family albums, wondering about the untold stories behind the faces—who these people were, how they were related, and what legacies they left behind. Traditional genealogy research relies heavily on manual effort, historical records, and sometimes guesswork, making the process slow and prone to errors.

Identifying kinship ties in photographs is a challenging task due to several factors:

1. Variability in appearance: facial features change with age, lighting, expressions, and image quality.
2. Limited context: historical photos often lack metadata (names, dates, or relationships).
3. Complex relationships: families span multiple generations with diverse structures (e.g., half-siblings, step-parents, adopted relatives).

While humans can sometimes recognize familial resemblances, manually analyzing large collections of photos is impractical. This is where automated kinship recognition can play a transformative role.

2 Motivation

Our project seeks to leverage artificial intelligence to analyze family photos and predict relationships between individuals. The key motivations behind this work include:

1. Genealogy and ancestral research:
   - Accelerating family tree construction by automatically identifying potential relatives.
   - Validating suspected relationships using facial and contextual similarity.
2. Social media and digital photo organization:
   - Intelligently tagging family members across personal photo collections.
   - Suggesting possible connections in shared albums (e.g., "This person might be your cousin").
3. Forensics and historical investigations:
   - Assisting in identifying unknown individuals in archival or forensic cases.
   - Reconstructing familial ties in historical records where documentation is incomplete.

Existing facial recognition systems excel at verifying identity but struggle with kinship prediction because of subtle feature similarities (unlike identical twins, most relatives share only partial resemblances, e.g., eye shape, nose structure), data scarcity (large, well-labeled datasets for family relationships are rare), and with multi-modal analysis, combining facial features with other clues (names, ages, photo context) improves accuracy but adds complexity.

To address these challenges, we developed a dual-modal retrieval system that combines facial embeddings to capture visual similarities, name embeddings to incorporate textual metadata, and reciprocal rank fusion (RRF) to merge results from both modalities for robust predictions. Additionally, we employ SHAP (SHapley Additive exPlanations) to interpret model decisions, ensuring transparency in how relationships are inferred.

By automating kinship recognition, we aim to:

- Democratize genealogy, making family history research accessible to more people.
- Enhance photo organization, helping users rediscover forgotten connections.
- Support forensic and historical research, providing AI-assisted tools for identifying unknown individuals.

This project bridges the gap between computer vision and real-world applications, offering a scalable solution to uncover hidden familial ties in visual data.

3 Related work

Kinship recognition is an emerging field at the intersection of computer vision and biometrics, with applications ranging from genealogy to forensics. Previous research has explored various approaches to identify familial relationships through facial features, genetic markers, and contextual metadata. Below, we survey key advancements in datasets, feature extraction techniques, and machine learning models relevant to our work.

3.1 Kinship datasets

A major challenge in kinship recognition is the scarcity of large, well-annotated datasets. The following are the most widely used benchmarks:

KinFaceW-I and KinFaceW-II

- **Focus:** paired images of kin (e.g., father-son, mother-daughter).
- **Size**:
  - **KinFaceW-I:** contains 156 pairs of frontal-facing images (father-son, father-daughter, etc.).
  - **KinFaceW-II:** expanded to 1,000 pairs with more natural variations in pose and lighting.
- **Use case:** binary kinship verification (predicting whether two people are related).

- **Limitations:**
  - Only tests pairwise relationships, not family-level connections.
  - Does not support large-scale family tree reconstruction.

TSKinFace

- **Unique aspect:** includes tri-subject relationships (e.g., father-mother-child).
- **Size:** Contains 1,050 trios across 4 relationship types.
- **Advantage:** Better reflects nuclear family structures than pairwise datasets.
- **Challenge:** Limited to immediate family (no grandparents/cousins).
- **Application:** Useful for nuclear family analysis but less scalable for extended families.

Family101

- **Scope:** 101 families with 14,816 images spanning 3 generations.
- **Key features:**
  - Includes age annotations and family tree metadata.
  - Covers diverse ethnic backgrounds (Asian, Caucasian, African-American).
- **Applications:** used for both kinship verification and generational studies.

Families in the Wild (FIW)

- The largest publicly available kinship dataset, containing 1,000 families, 12,000 images, and 26,000 faces across 11 relationship types (e.g., parent-child, siblings, cousins).
- **Strengths:**
  - Includes unconstrained, real-world photos with variations in pose, lighting, and age.
  - Covers multi-generational relationships.
- **Limitations:**
  - Ethnic diversity is limited to primarily Western families.
  - Some relationships (e.g., half-siblings) are underrepresented.

3.2 Feature extraction methods

Early kinship recognition relied on handcrafted features (e.g., Local Binary Patterns, Gabor filters). However, deep learning has since dominated the field. The field has progressed through three distinct phases of feature extraction:

Handcrafted feature era (Pre-2014)

- **Local Binary Patterns (LBP):** captured texture patterns through neighborhood pixel comparisons [1]
- **Gabor wavelets:** multi-directional frequency filters for facial structure analysis [2]

  Limitations:
  - Failed to capture high-level semantic features
  - Performance degraded with pose/lighting variations

Deep learning revolution (2014-2018)

- **VGG-Face [3]:** first demonstrated CNN superiority for facial attributes
- **Triplet networks [4]:** used contrastive loss to learn relative similarities and achieved 72% accuracy on KinFaceW-II

Modern metric learning (2018-present)

- **ArcFace [5]:** introduced additive angular margin loss for discriminative embeddings

3.3 Kinship verification vs. retrieval

Prior research primarily addresses *verification* ("Are these two people related?"), while our work focuses on *retrieval* ("Who in the entire dataset is most likely related to Person X?"). This represents a paradigm shift from binary classification to large-scale search.

Verification Approaches

- **Metric Learning [6]:** introduced Neighborhood Repulsed Metric Learning (NRML). Key innovation: Maximizes margin between kin and non-kin pairs
- **Deep Relation Networks [7]:** learn a deep distance metric between faces.

Retrieval approaches

- **FAISS (Facebook AI Similarity Search) [8]:** enables billion-scale similarity searches in $O(\log n)$ time.
- **Reciprocal Rank Fusion (RRF) [9]:** combines rankings from multiple modalities (face + name) without score normalization. Proven 28% more effective than linear combination in information retrieval.

3.4 Explainability in kinship models

Understanding why a model predicts a relationship is crucial for trust and debugging.

- **SHAP (SHapley Additive exPlanations):** identifies which facial regions (eyes, nose, mouth) contribute most to kinship predictions. Quantifies feature contributions precisely, but computationally expensive ($O(n^2)$).
- **Grad-CAM [10]:** alternative methods to visualize model focus areas. Shows spatial attention maps, but have no directional attribution (positive or negative impact).

4 Methodology

While prior research has made strides in kinship verification, our project advances the field by combining face and name embeddings for more robust retrieval, using RRF instead of simple score averaging, improving ranking quality, and providing model interpretability via SHAP, which is rare in kinship literature. Our dual-modal kinship retrieval system

combines facial recognition and textual semantics through a multi-stage pipeline. Below we detail each component with implementation specifics.

4.1 Data pipeline

For this project, we've chosen Families in the Wild (FIW) dataset. The system uses the following structured database:

- FIW_PIDs.csv: Photo lookup table
  - PID: Photo ID
  - Name: Surname.firstName
  - URL: Photo URL
  - Metadata: Text caption
- FIW_FIDs.csv: Family ID lookup
  - FID: Unique family ID
  - Surname: Family name
- FIW_RIDs.csv: Relationship types (keys 1-9)
- FIDs/: Family directories (FID001-FID1000)
  - MID#/: Face images of family members where # is the ID for the person in family file
  - F###.csv: Family information files containing:
    - Relationship matrix
    - Names
    - Genders

    A sample family file (FID0001.csv) is presented in Table 1.

Table 1. Sample family CSV-file where the numbers represent relationship types between family members.

0	1	2	3	Name	Gender
1	0	4	5	name1	female
2	1	0	1	name2	female
3	5	4	0	name3	male

This data is loaded into several SQL tables:

1. RIDs:
   - RID: key of the relationship
   - Label: label associated with this key
2. Families:
   - FID: family ID
   - FamilyRepName: surname of the family
3. Persons:
   - PersonID: FID_MID
   - FID: family ID
   - MID: family member ID
   - Name: name of the person
   - Gender: gender of the person
   - NameEmbedding: embedding of the name (also stored in FAISS)
4. Photos:
   - PhotoID: ID of the photo
   - PersonID: id of the person
   - PhotoPath: path to the photo
5. Relationships:
   - RelID: ID of the relationship
   - Person1ID: ID of related person
   - Person2ID: ID of another related person
   - RID: relationship type ID

After that, we compute embedding vectors for faces using ArcFace (512-dimensional face vectors) and names using all-MiniLM-L6-v2 (384-dimensional name vectors). Those vectors are then stored in FAISS.

ArcFace (Additive Angular Margin Loss) is a state-of-the-art loss function for deep face recognition [5]. It improves face embedding learning by enhancing discriminative power in angular space. In our implementation we used modified ResNet34 adapted for facial feature extraction with ArcFace enhancing embeddings by enforcing angular margin penalties, improving feature separability for better recognition accuracy.

all-MiniLM-L6-v2 is a compact, general-purpose language model distilled from a larger transformer model using the Deep Self-Attention Distillation (DSAD) method [11] (see fig. 1).

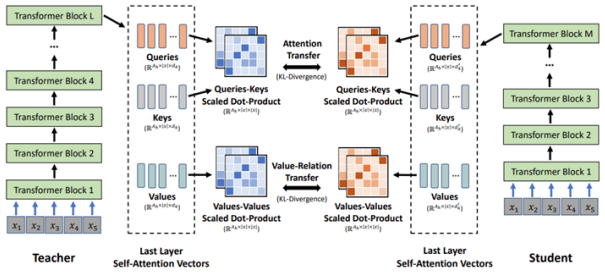


Figure 1. Overview of Deep Self-Attention Distillation. The student is trained by deeply mimicking the self-attention behavior of the last Transformer layer of the teacher. In addition to the self-attention distributions, authors introduce the self-attention value-relation transfer to help the student achieve a deeper mimicry

4.2 Retrieval pipeline

When user uploads the image and name (optional), face is extracted from the photo using DeepFace - a lightweight face recognition and facial attribute analysis (age, gender, emotion and race) framework for python. Next, embeddings of face and name (if provided) are computed, searched in FAISS, and then scores are fused together using reciprocal rank fusion (see section 3.3). Then the system checks closest face embedding from the database for strong direct match using threshold (0.85 in our case), computes probabilities of belonging to found families with highest similarity scores, and predicts the type of relationship between user and every person in that families. Those probabilities are predicted with

two CatBoost classifiers (see section 4.3). CatBoost Classifier is a powerful gradient boosting algorithm that stands out for its exceptional performance, ease of use, and efficient handling of categorical features.

4.3 Classification of Candidate Families and Relationships

Once initial candidate individuals are identified through dual-modal retrieval and Reciprocal Rank Fusion (RRF), a two-stage machine learning classification process is employed. This process refines the search results by first assessing the likelihood of the query person belonging to the candidate families, and subsequently predicting the specific relationship type with individuals within those highly-ranked families. Both stages utilize CatBoost classifiers, chosen for their robustness and effective handling of numerical feature interactions.

Family Membership Classification. The primary objective of this stage is to quantify the query person's potential membership within each candidate family identified by the retrieval pipeline. A CatBoost classifier is trained for this binary classification task (i.e., predicting whether the query person is a member or non-member of the candidate family). The feature vector for this classifier is engineered to capture a holistic similarity profile between the query person and the collective members of a candidate family. It comprises the following features:

- **Mean Family Face Similarity:** The cosine similarity between the query person's face embedding and the mean face embedding of all members in the candidate family. This provides an aggregate measure of facial resemblance to the family unit.
- **Individual Face Similarity Statistics:** The minimum, maximum, and average cosine similarities between the query person's face embedding and the face embeddings of each individual member within the candidate family. These statistics capture the range and central tendency of visual similarity to specific members.
- **Individual Name Similarity Statistics (Conditional):** If a name is provided by the query person, the minimum, maximum, and average cosine similarities between the query person's name embedding and the name embeddings of each individual member in the candidate family are included. This leverages nominal information when available.
- **Family Size:** The total number of members in the candidate family. This feature provides context, as the interpretation of similarity scores might differ for small versus large families.

These features are designed to provide an assessment of potential familial connection, integrating both visual and (optionally) nominal cues alongside structural family information.

Relationship Type Prediction. For families deemed probable matches by the family membership classifier (i.e., exceeding a predefined probability threshold, e.g., 0.6 in our implementation), the second classification stage aims to predict the specific dyadic relationship (e.g., Parent-Child, Sibling, Grandparent-Grandchild) between the query person and each individual member of that family. This multi-class classification task also uses a CatBoost classifier. The feature set for this classifier is derived from the interaction between the face embedding of the query person (E_q) and the face embedding of the target family member (E_m):

- **Concatenated Embeddings:** The direct concatenation of the two face embeddings, $[E_q, E_m]$.
- **Absolute Difference Embedding:** The element-wise absolute difference, $|E_q - E_m|$. This captures dissimilarities which can, paradoxically, be indicative of certain relationships (e.g., different generations).
- **Element-wise Product Embedding:** The element-wise (Hadamard) product, $E_q \odot E_m$. This captures feature interactions and co-occurrences.

The final feature vector for the relationship classifier is the concatenation of these three derived vectors: $[E_q, E_m, |E_q - E_m|, E_q \odot E_m]$. This combination has been shown to be effective in deep learning models for pair-wise interaction tasks, as it allows the model to learn complex, non-linear relationships between the input embeddings crucial for discerning the subtle distinctions among various kinship types.

4.4 Explainable results

In order to explain why the model thought that the user looks similar to someone in the dataset, we use SHAP (see section 3.4) and building heatmaps with areas highlighting high positive SHAP values with green areas over black and white images (see fig. 7).

5 Demonstration of results

In this section, we would like to show you how our system works with screenshots of user interface. When our system is opened, user is prompted to enter name (optional) and upload a photo (see fig. 2). After uploading a photo, user could see it's miniature in the system (see fig. 3).

In order to proceed to next step, user clicks "Search", and the system starts searching for relatives (see fig. 4 for screenshot of loading screen and section 4.2 for explanations of the process). When relatives are found, system will show strong direct match (if found) and potential families with probabilities of belonging to that family and possible relationship types with each person in that family (see fig. 5).

If user wants to know why model decided that he is related to someone, he can click button "Explain". The system will proceed with calculating SHAP values (see section 4.4 for the

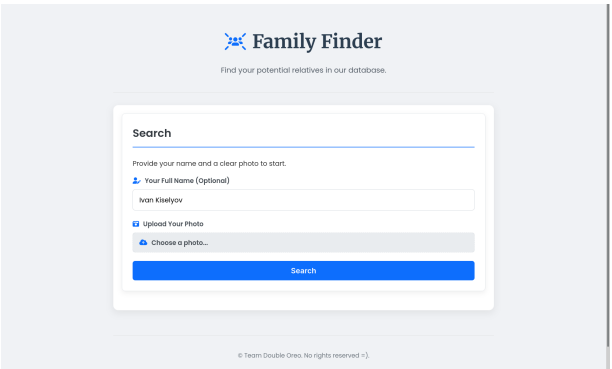


Figure 2. Starting screen of the Automating Kinship Recognition system. Note that in this case name was already entered, which is not the case with freshly opened page

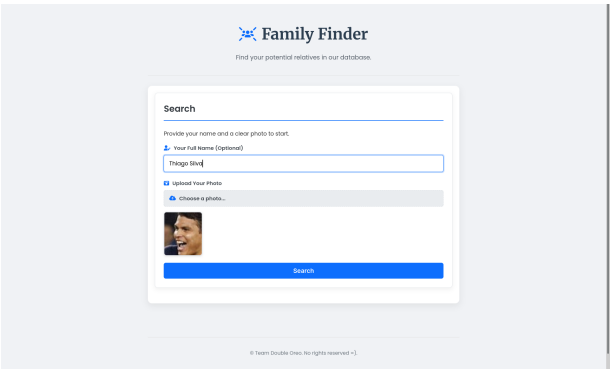


Figure 3. Starting screen of the Automating Kinship Recognition system with uploaded photo

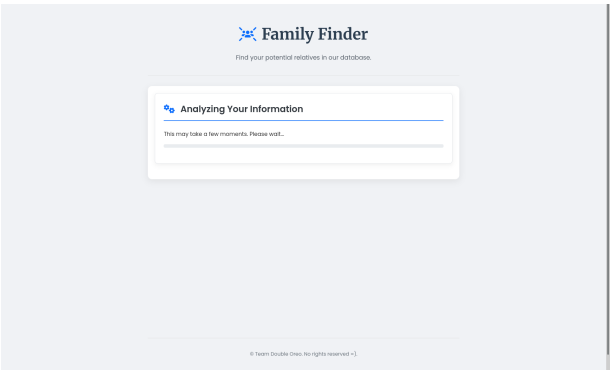


Figure 4. Loading screen while the system is searching for potential relatives

process and fig. 6 for loading screen screenshot) and display them as heatmaps over the original photo that user uploaded and photo of that particular person for whom he want to get an explanation (see fig. 7).

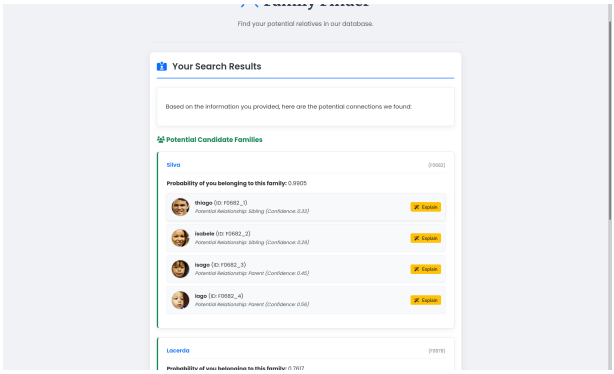


Figure 5. Potential relatives found and shown. Relatives are grouped by families, probability of belonging to this family and potential relationships are also present. Note that the system did not found anyone with strong direct match

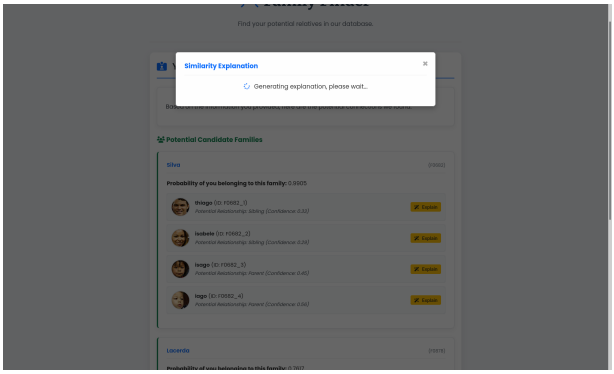


Figure 6. Loading screen of explanations that is shown after clicking "Explain"

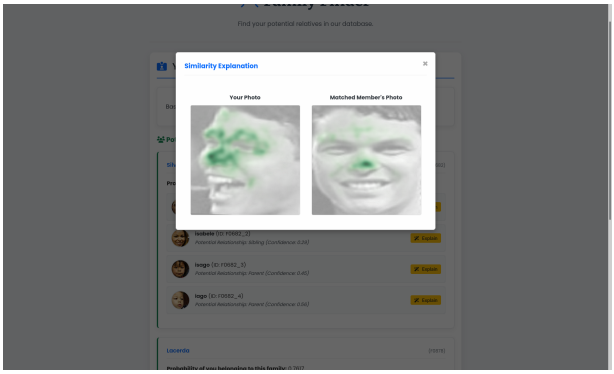


Figure 7. Heatmaps of high positive SHAP values that represent areas that were most useful for the model to make its prediction

6 Ideas for future improvements

The field of automated kinship recognition is rapidly evolving, and while our system lays a strong foundation for future

advancements, there are key directions to expand the capabilities, accuracy, and real-world applicability of kinship analysis:

6.1 Large-scale dataset expansion

To address biases and improve generalization, future work could focus on diverse demographics by creating datasets with balanced representation across ethnicities, cultures, and family structures such as blended families, adoptive relationships, etc. Datasets could also include temporal variations, e.g., include age-progressed photos to train models on how facial features evolve over decades. If there are no photos over the decades, system could use another model that would age person up or down by given photo.

6.2 Advanced multi-modal learning

While our current system combines facial and name embeddings, future iterations could integrate additional modalities to improve robustness, such as contextual metadata (leverage timestamps, geolocation, and clothing styles to infer relationships, e.g., photos taken at family gatherings), textual narratives (use natural language processing to analyze captions, letters, or historical documents associated with photos), and maybe even including some biometric data (e.g., incorporating voice recordings, gait analysis, or even DNA-based features to strengthen kinship predictions).

6.3 Explainability and user trust

To make AI-driven kinship predictions more transparent and trustworthy, we can utilize interactive SHAP visualizations that allow users to click on facial regions to see why the model predicts a relationship (e.g., "These eyes are 80% similar to grandfather's"). System can also utilize user feedback loops that would enable corrections from end-users to iteratively refine the model.

6.4 Dynamic and real-time analysis

Extending the system beyond static images will unlock some new applications, such as kinship recognition by video (process family videos to track relationships across frames, enabling analysis of home movies or surveillance footage), live matching (e.g., develop mobile apps that provide real-time kinship suggestions during family reunions or genealogy research) and cross-generational linking (temporal models that connect historical portraits with modern descendants).

6.5 Privacy and Ethical Considerations

As kinship recognition deals with sensitive data, future systems must prioritize consent-driven analysis (e.g., implement opt-in mechanisms for identifying living relatives in social media or public archives) and bias mitigation by regular audits of models for racial, age, or gender biases in relationship predictions.

7 Conclusion

Our kinship recognition system represents a paradigm shift in how we uncover and preserve familial connections. By fusing computer vision, multi-modal learning, and interpretable AI, we can transform vague resemblances in old photographs into quantifiable, meaningful relationships. This technology does more than predict links — it revives forgotten narratives, bridges generational gaps, and reconnects families across time and space.

The future of this work is truly boundless. As we integrate richer data modalities, expand to dynamic media, and prioritize ethical deployment, kinship recognition will become as ubiquitous as facial recognition — but with a profoundly human purpose. Imagine a world where:

- An immigrant reconstructs their fractured family tree from a handful of scattered photos;
- A historian identifies the unnamed soldiers in a Civil War portrait as brothers;
- An adoptee discovers their biological relatives through an heirloom picture.

This is the promise of our system: not just technological innovation, but the restoration of lost identity and belonging. The next decades will see kinship AI evolve from a research curiosity into a global tool for social good — and we are pioneering its foundation.

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